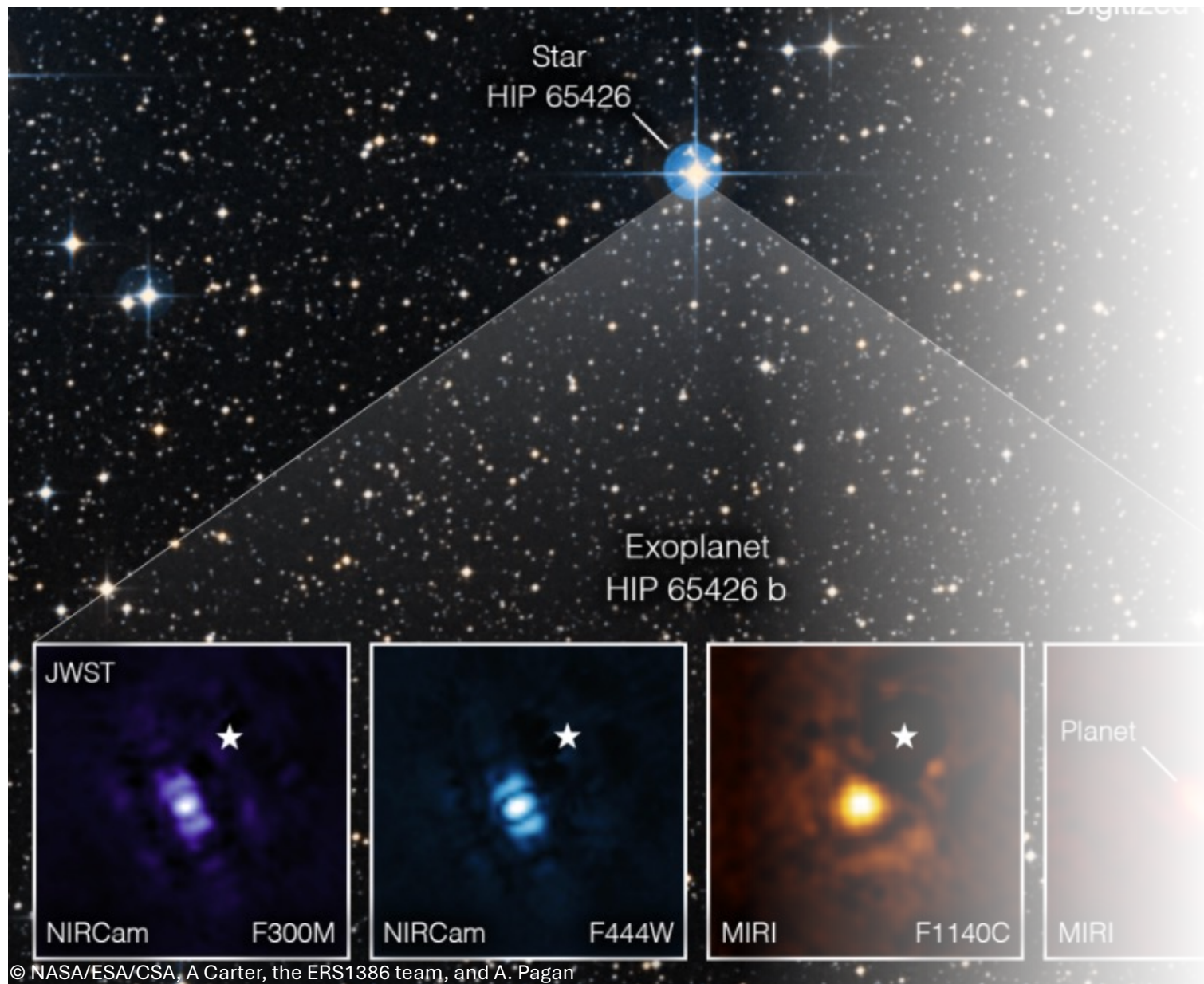




JWST/ NIRCam, MIRI Coronagraphy Data Reduction and Analysis Tutorial

Taichi Uyama
(Astrobiology Center)

2026/6/2-4

JWST data reduction tutorial lecture

Outline

Introduction

- Exoplanetary science
- High-contrast imaging studies

Method

- PSF subtraction technique

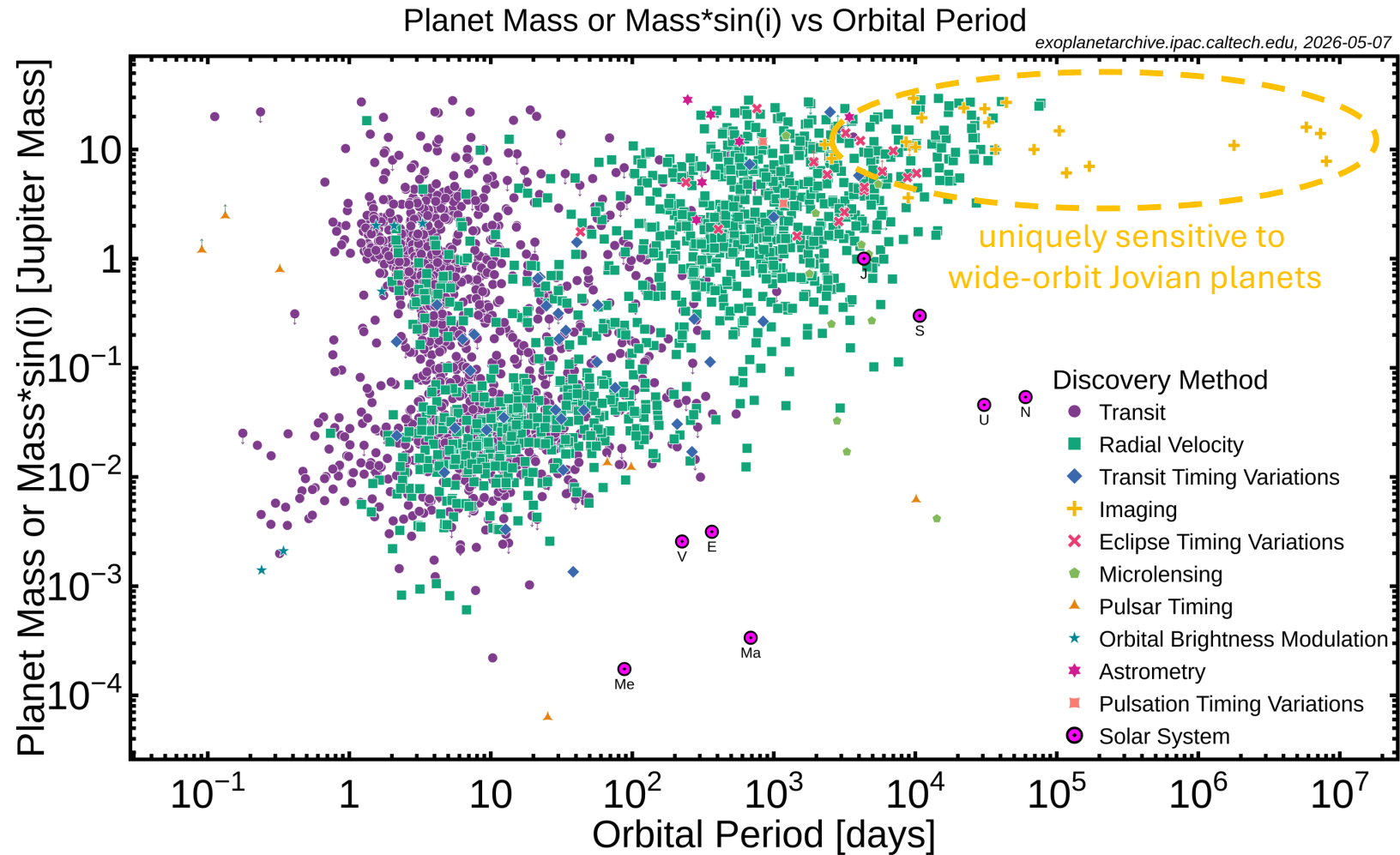
JWST Instruments

- NIRCам and MIRI Coronagraphy

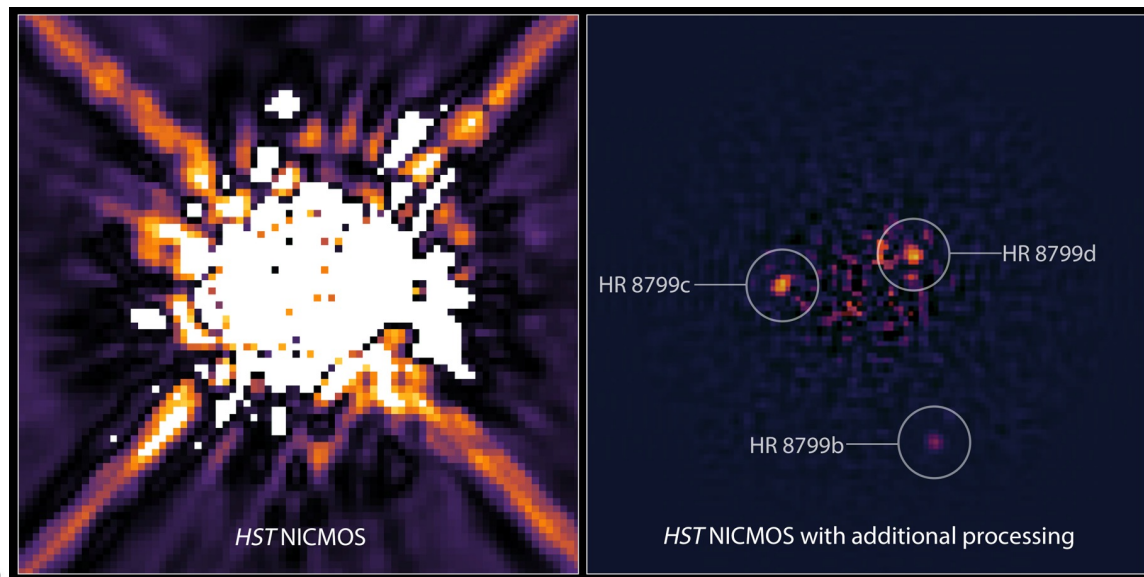
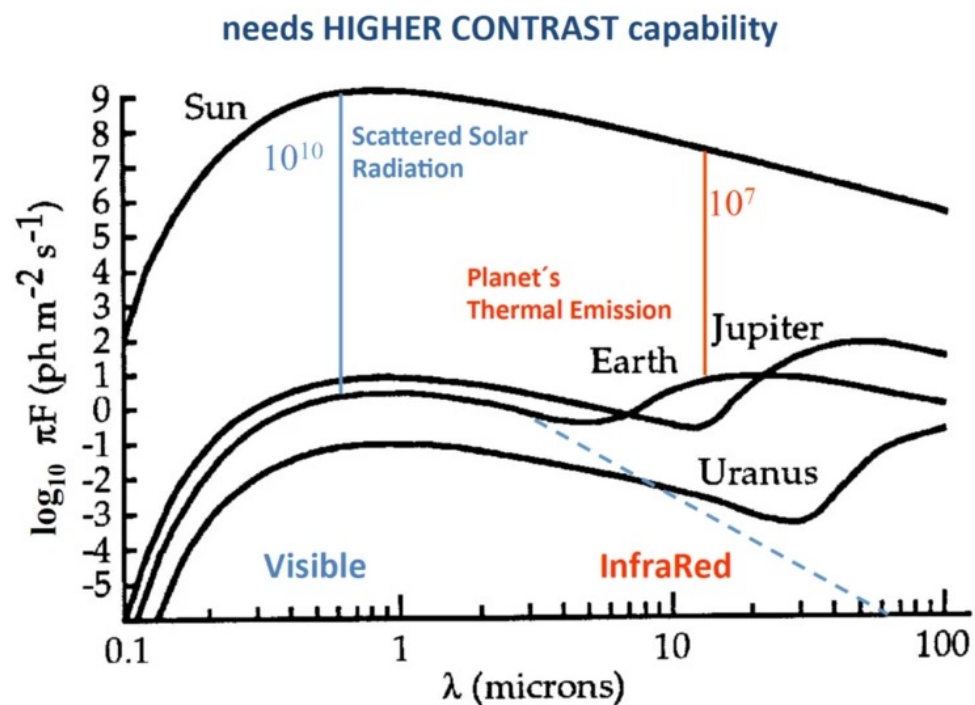
Data Reduction

- Pre-processing with JWST pipeline and STPSF
- Post-processing with SpaceKLIP
- Data analysis with SpaceKLIP

Direct (High-contrast) Imaging of Exoplanets



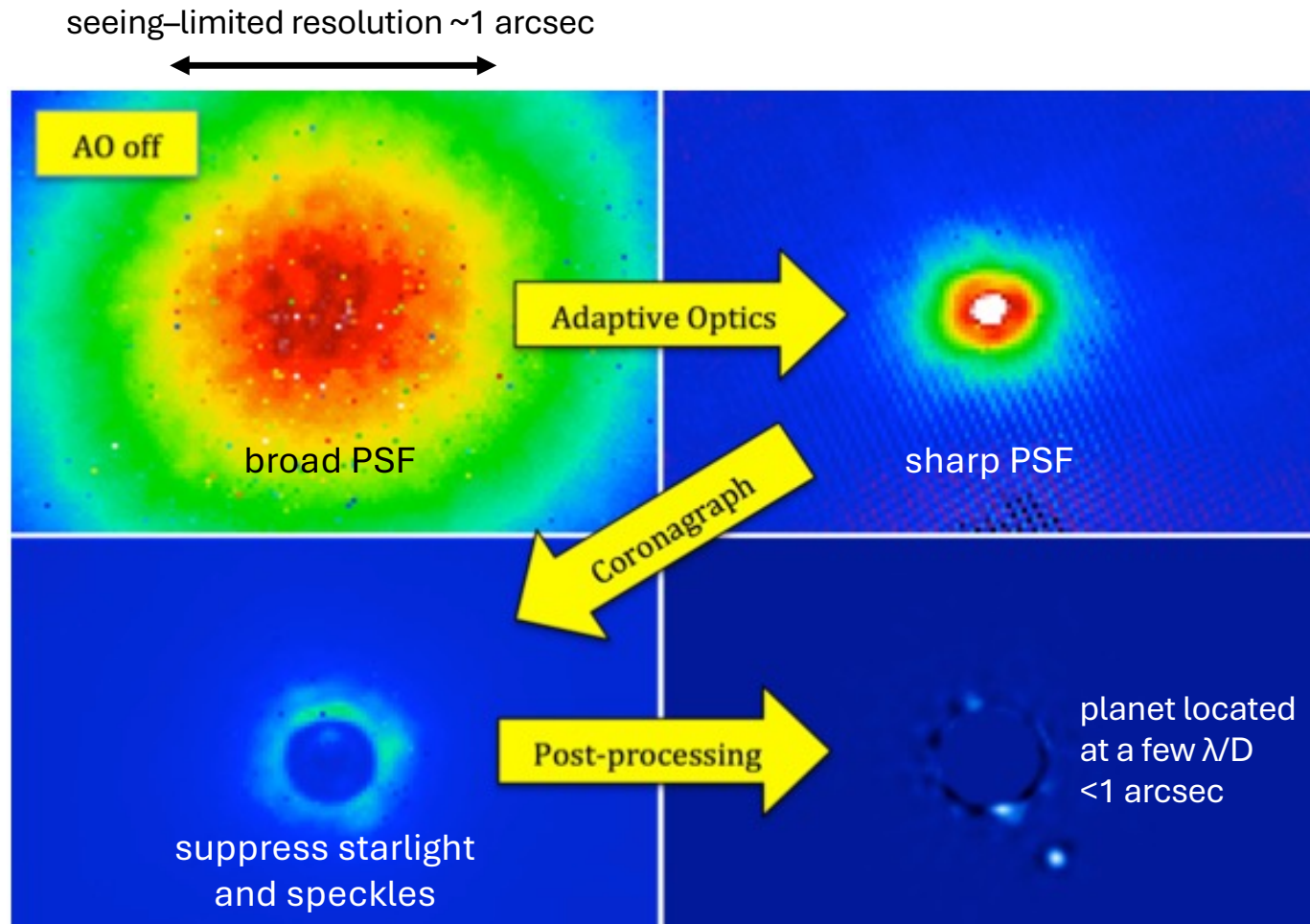
What is 'High-contrast'?



HR 8799 system imaged by HST/NICMOS in 1998 (©NASA)

Comparison of spectra between the Sun (hosting star) and Solar-system planets. (The Way forward; Fridlund et al. 2016)

Flowchart of High-contrast Imaging

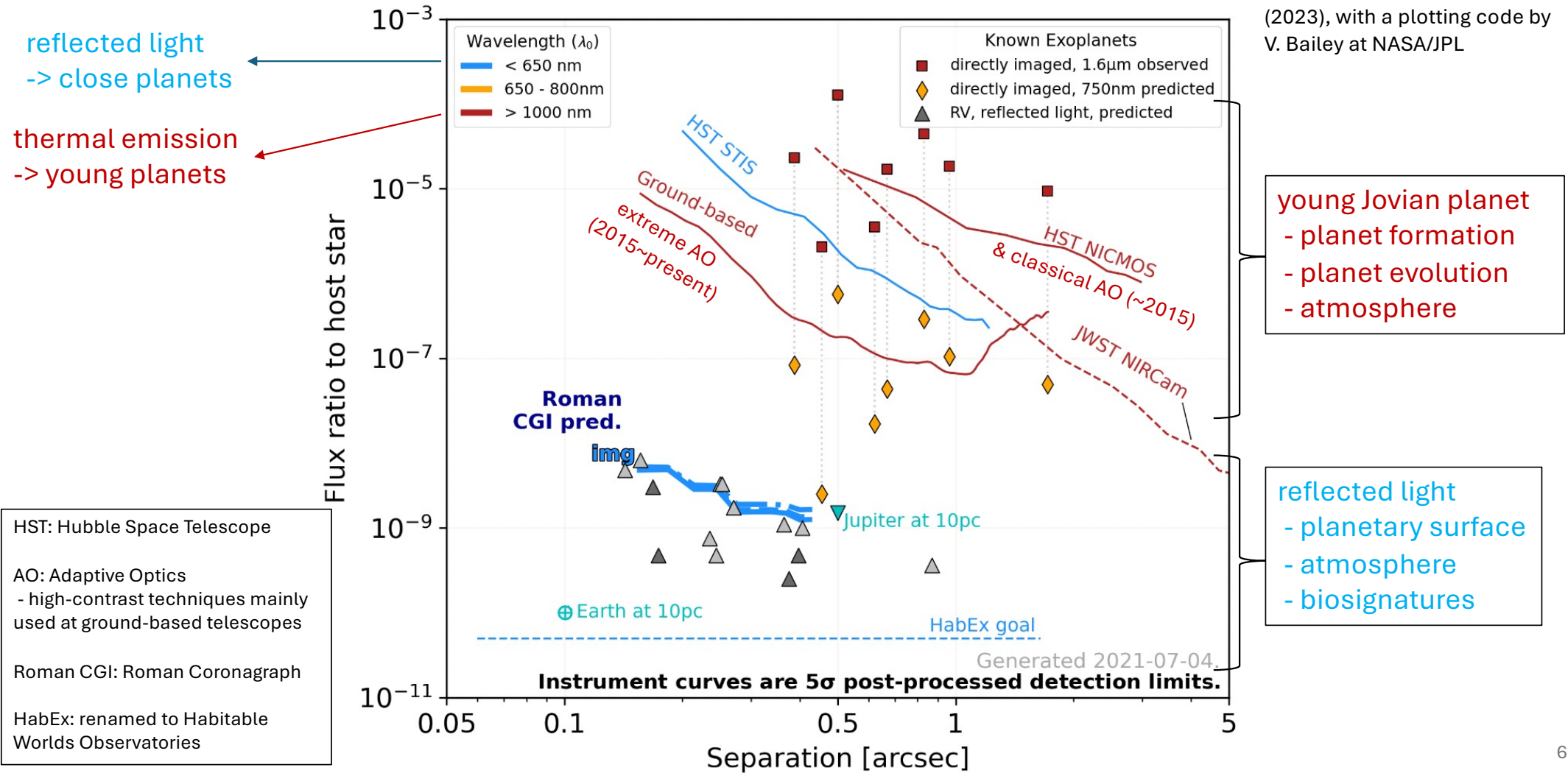


Adaptive Optics
=
Wavefront Sensing
(including JWST)
+
Wavefront Correction
(ground-based,
Roman, HWO)

In an ideal case, AO
provides diffraction-
limited PSF
(FWHM $\sim \lambda/D$).

Schematic flowchart for a ground-based telescope ©C. Marois

High-contrast Capabilities and Related Science



Outline

Introduction

- Exoplanetary science
- High-contrast imaging studies

Method

- PSF subtraction technique

JWST Instruments

- NIRCам and MIRI Coronagraphy

Data Reduction

- Pre-processing with JWST pipeline and STPSF
- Post-processing with SpaceKLIP
- Data analysis with SpaceKLIP

Calibration, Reduction, and Analysis Steps

the pipeline
almost
customized
these steps for
coronagraph
data

skipped in this
lecture
(details will be
presented in
the 2nd
workshop)

the main
contents of
this lecture

1. Calibration, Pre-processing

1. Dark/sky subtraction, flat-fielding, distortion correction,
badpix masking

2. (Spectro)photometric calibration

3. Image alignment

2. Post-processing

1. PSF subtraction

3. Analysis

1. Forward modeling

2. Calculate contrast limit

3. Extract spectrum/photometry

Difficulty Level

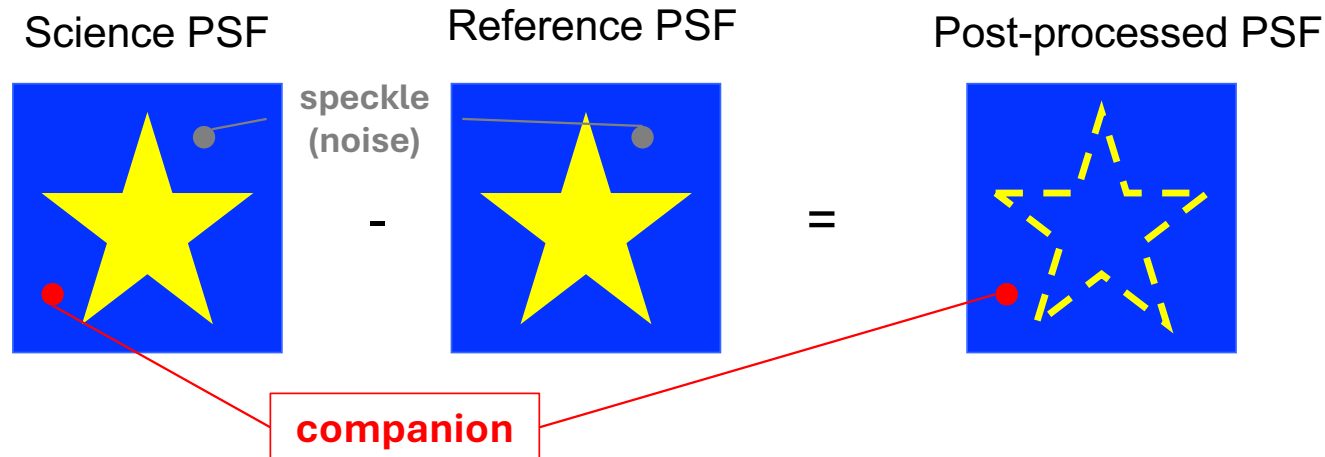
Basic

Medium

Advanced

Post-processing - Differential Imaging

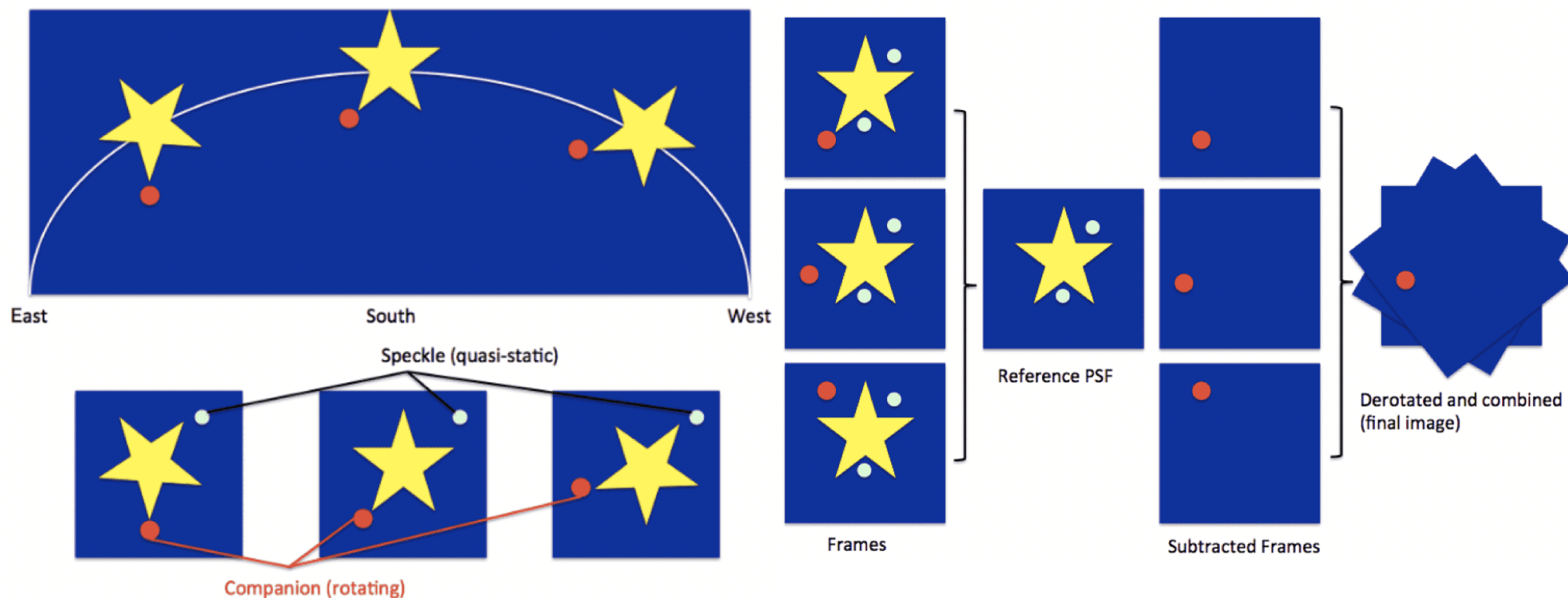
- Produce the most likely reference PSF to the science PSF
- Reference PSFs can be made by differential imaging technique
 - *Angular Differential Imaging (ADI)* – using field rotation
 - *Reference-star Differential Imaging (RDI)* – observing a PSF reference star
 - *Spectral Differential Imaging (SDI)* – $\rho_{\text{speckle}} \propto \lambda$, or searching for specific lines and more..



Angular Differential Imaging

<- this color+underbar has a link

- Widely-used for high-contrast imaging data
- by fixing pupil (allowing background rotation), one can generate a reference PSF consisting of a stellar PSF and speckles
- field rotation is essential, otherwise the real source would suffer from ‘self-subtraction’



Outline

Introduction

- Exoplanetary science
- High-contrast imaging studies

Method

- PSF subtraction technique

JWST Instruments

- NIRCам and MIRI Coronagraphy

Data Reduction

- Pre-processing with JWST pipeline and STPSF
- Post-processing with SpaceKLIP
- Data analysis with SpaceKLIP

JWST Coronagraphy: Pros and Cons

Pros

- Higher sensitivity and diffraction-limited angular resolution
- Great instrumental stability and wavelength coverage (particularly $\lambda > 3\mu\text{m}$)

Cons

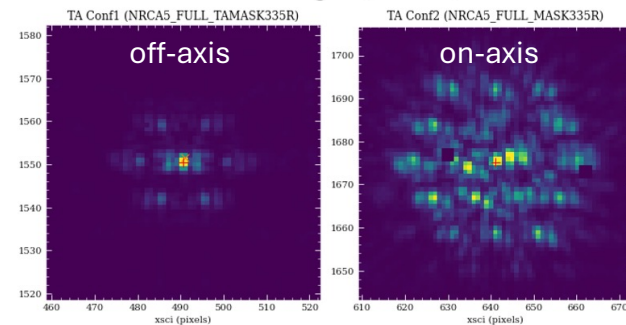
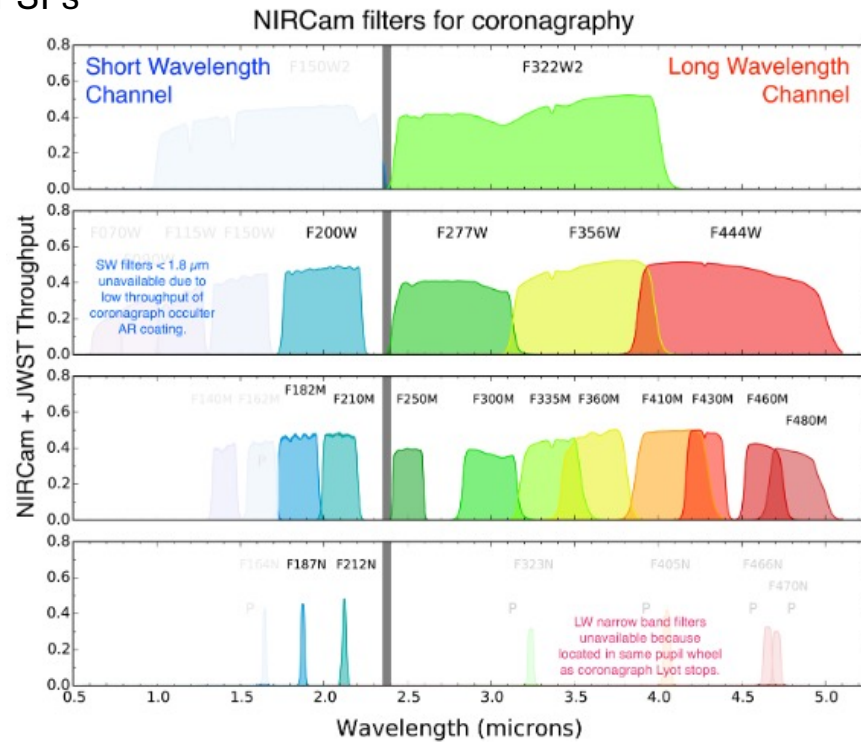
- Coronagraph is not as good as ground-based telescopes (larger IWA)
- Observational limitations are stricter than ground-based telescopes
 - due to the slow telescope slew, observing a reference star increases overheads
- Higher oversubscription ratio (JWST cycle 5 was >10) than other telescopes

if you are interested in proposing JWST observations,
check [JWST High-contrast imaging roadmap in the user documentation](#)

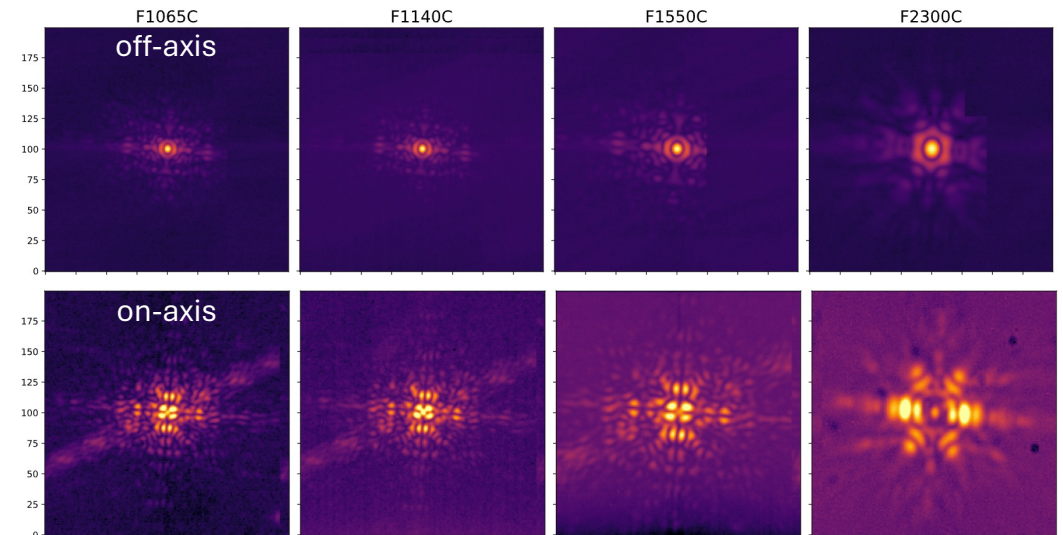
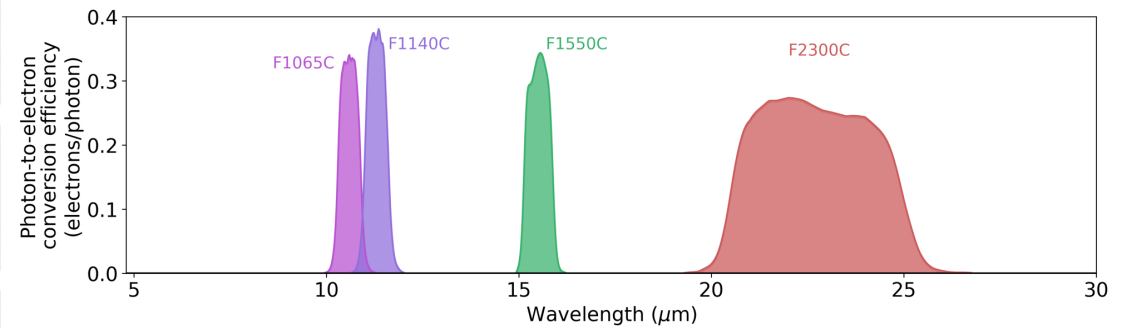
NIRCam and MIRI Configurations for Coronagraphy

top: filters

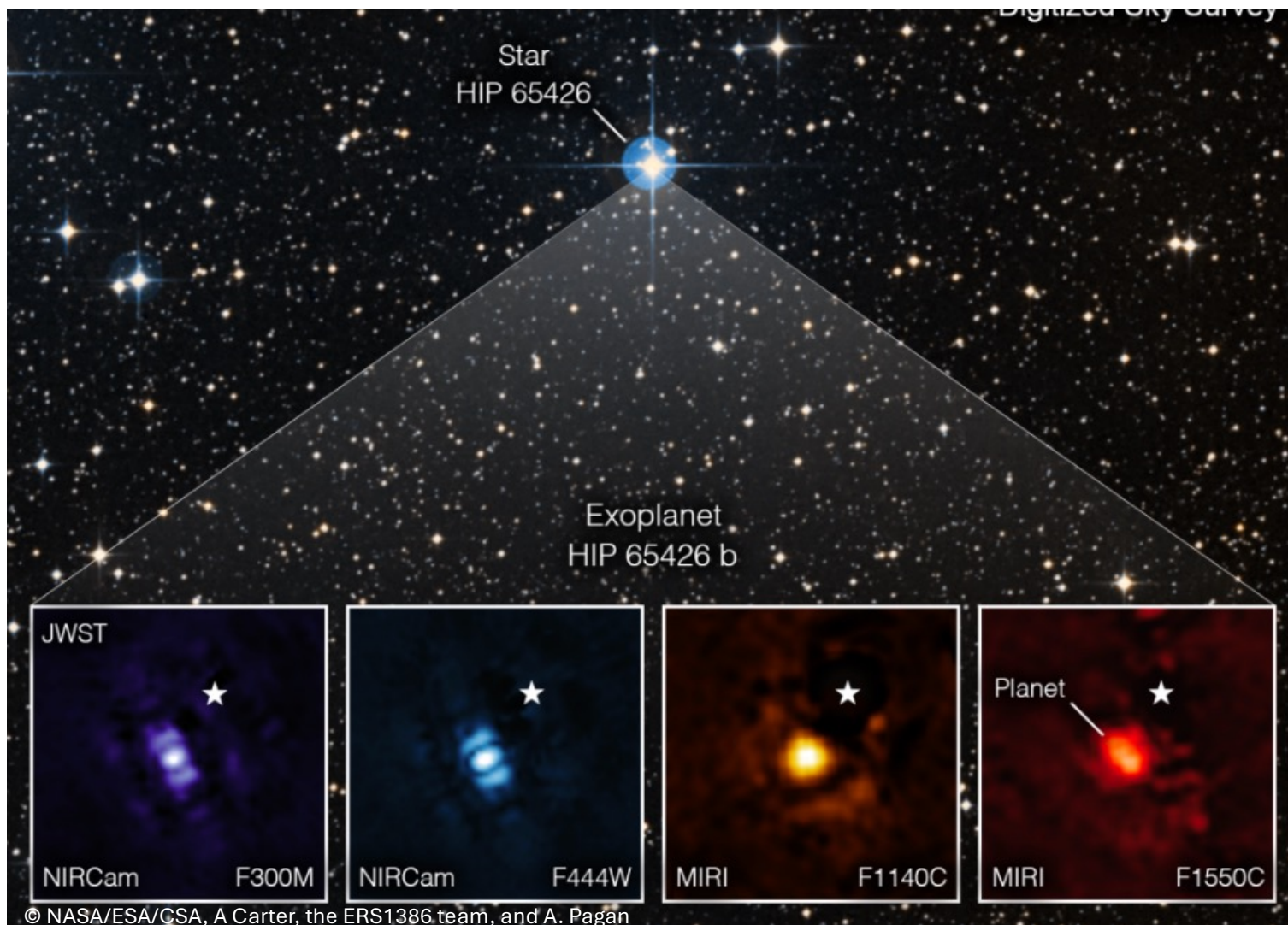
bottom: PSFs



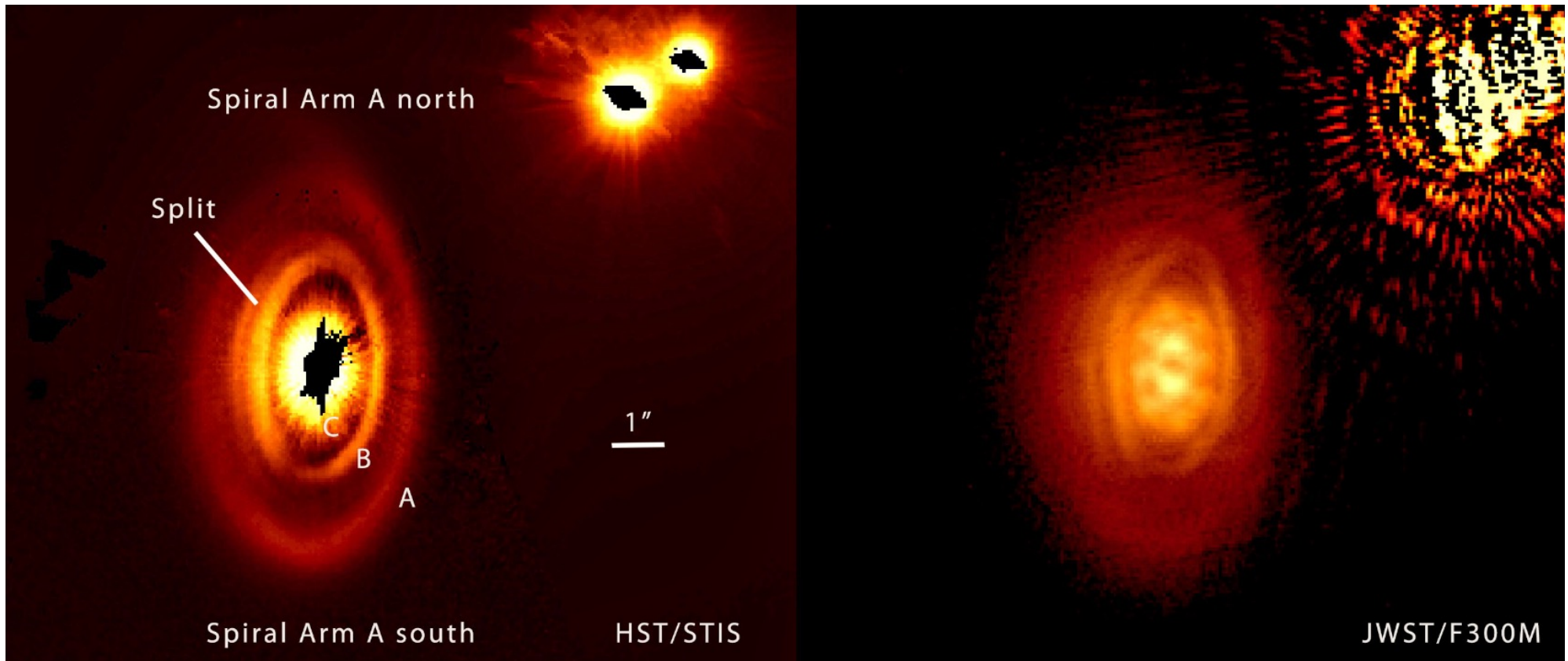
MIRI filters for coronagraphy



Example (exoplanet): HIP 65426b

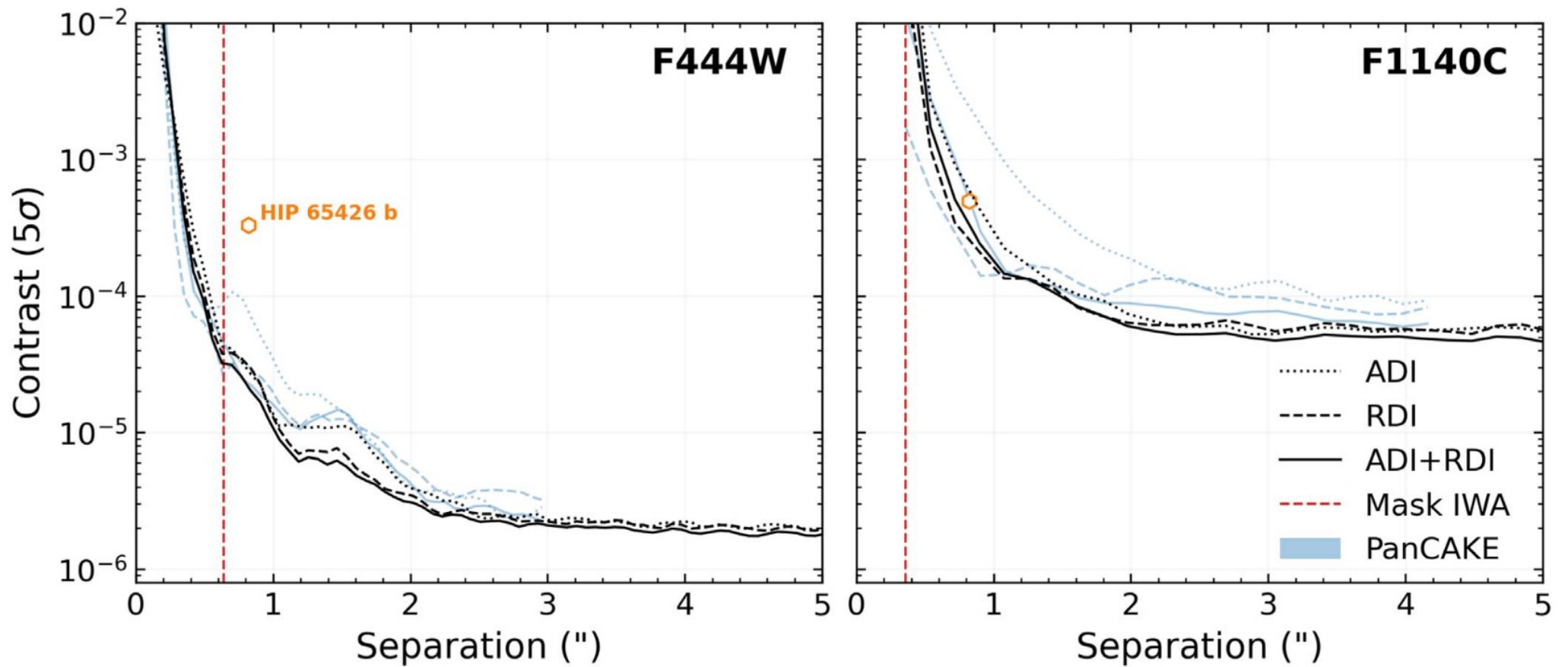


Example (disk) : HD 141569A



Millar-Blanchaer et al. 2025

NIRCam and MIRI High-contrast Performance



Carter et al. 2023

Outline

Introduction

- Exoplanetary science
- High-contrast imaging studies

Method

- PSF subtraction technique

JWST Instruments

- NIRCам and MIRI Coronagraphy

Data Reduction

- Pre-processing with JWST pipeline and STPSF
- Post-processing with SpaceKLIP
- Data analysis with SpaceKLIP

SpaceKLIP Calibration/Reduction Steps

0. Reformatting fits file (Stage 0-> Stage 1)

※ Configurations of the JWST pipeline parameters are empirically customized

1. Calibration/Pre-processing with JWST pipeline (Stage 1 -> Stage 2)

1. Background subtraction, flat-fielding, distortion correction
2. Photometric calibration

2. Post-processing (Stage 2 -> Stage 3)

1. Badpix masking , image (re-)alignment with STPSF
2. PSF subtraction

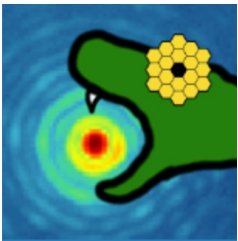
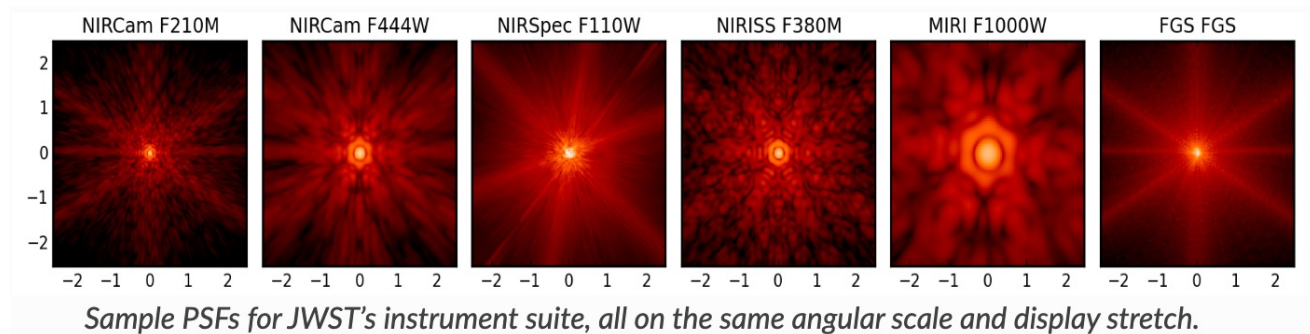
3. Analysis

1. Calculate contrast limit
2. Extract photometry



Useful Public Pipelines/Libraries

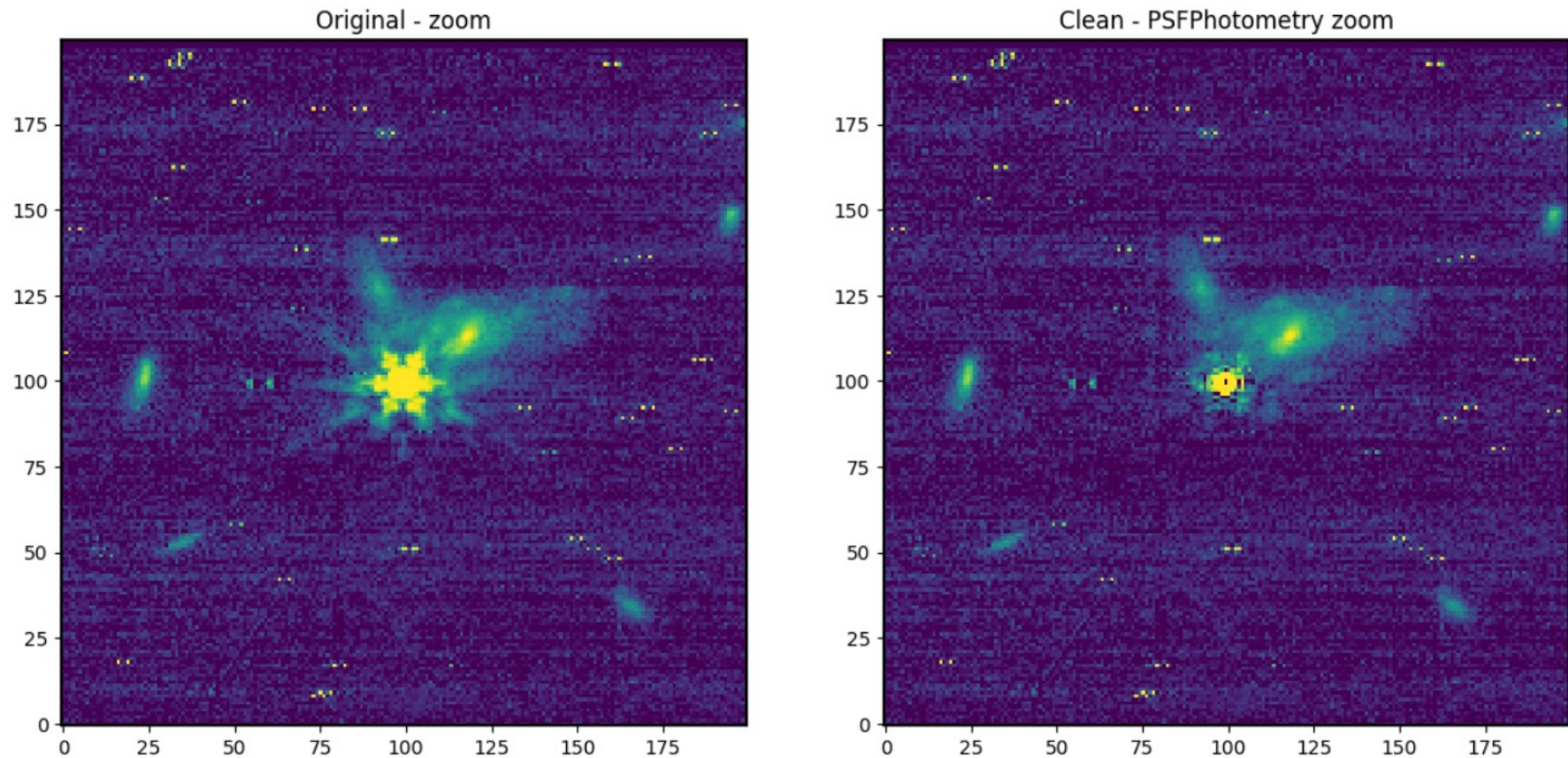
- [JWST pipeline](#)
 - Stage 0 (raw) -> Stage 2 (pre-processed)
- [STPSF](#) (previously WebbPSF)
 - Simulating on-axis and off-axis PSFs based on telemetry data



- [SpaceKLIP](#) (wrapping JWST/STPSF)
 - Stage 0 -> Stage 3 (post-processed)
 - Data analysis using stage 3

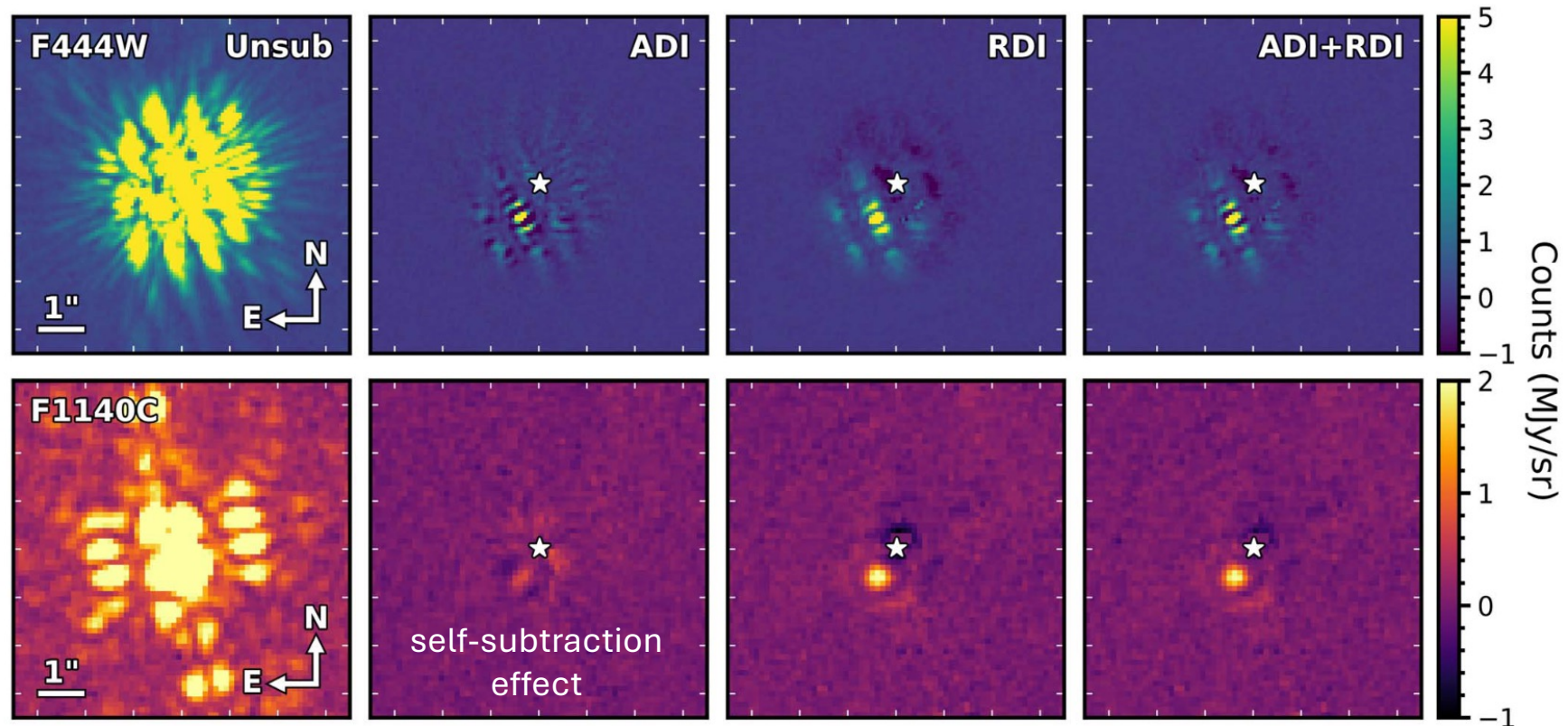
PSF Subtraction with Simulated PSF

- for conservative PSF subtraction, STPSF works to some extent
- for high-contrast, we need real reference data taken at the same epoch



What Reference Data (post-processing method) is Good?

- generally, RDI is the best for JWST data (but needs a PSF reference star)
- ADI is available but the field rotation is limited to ~ 10 deg
- SDI is also doable with 2+ filters, but technically difficult for JWST data.



Example images of HIP 62546b; Carter et al. 2023

For the Hands-on Session with SpaceKLIP

goal 0 (preparation) : Prepare the JWST data by the start of the hands-on session

goal 1 (Stage 0 -> 3): Complete post-processing

goal 2 (analysis) : Obtain contrast curves and extract photometry/astrometry

TUTORIALS

Tutorial Notebooks

- goal 1 ☒ Tutorial for NIRCcam Coronagraphy Reduction with spaceKLIP
- goal 1 Tutorial for MIRI Coronagraphy Reduction with spaceKLIP
- goal 2 Tutorial for NIRCcam Post-Pipeline Contrast Analyses Using spaceKLIP
- goal 2 Tutorial for MIRI Post-Pipeline Contrast Analyses Using spaceKLIP
- goal 0 + α MAST query and download tools for coronagraphic datasets

Tips for Hands-on Session

- First use the Early-Release Science (ERS) data; **HIP 65426 (Carter et al. 2023)**
 - you need the corresponding PSF reference data set for RDI (can be downloaded with the same proposal ID as the science data)
 - the MIRI data sets are much heavier than the NIRCам data sets (downloading and running spaceKLIP takes more time)
- Confirm that you successfully obtain the post-processed images (e.g. ADI/RDI/ADI+RDI) and analysis outputs (contrast curves, extracted photometry).
- Then you may try to modify some parameters (particularly [post-processing parameters in pyKLIP](#)) or try different data sets from archive.
 - refer to the list of peer-reviewed papers for the data profiles.
 - companion data are simpler than disk data.
- Some studies adopted special reduction/analysis steps or didn't observe (or failed acquiring) PSF reference stars. You may avoid them.

SpaceKLIP Implements MAST Data Query

- you don't need to use the complicated MAST platform or MAST API
- but downloading requires MAST API TOKEN

```
[1]: import spaceKLIP

table = spaceKLIP.mast.query_coron_datasets('NIRCam', 'F356W')

print(f"Found {len(table)} total rows. Here's the first few:")
table[:5]
```

Found 34 total rows. Here's the first few:

[1]: Table length=5

visit_id	start time	filter	coronmsk	targname	obslabel	duration	numdthpt	program	title	pi_name
str12	str16	str5	str9	str22	str54	float64	int64	int64	str75	str20
V01386001001	2022-07-29 22:01	F356W	MASKA335R	* phi Cen	NIRCam 335R - REF	83.426	9	1386	High Contrast Imaging of Exoplanets and Exoplanetary Systems with JWST	Hinkley, Sasha
V01386002001	2022-07-30 01:13	F356W	MASKA335R	HD 116434	NIRCam 335R - Roll 1	617.946	1	1386	High Contrast Imaging of Exoplanets and Exoplanetary Systems with JWST	Hinkley, Sasha
V01386003001	2022-07-30 03:16	F356W	MASKA335R	HD 116434	NIRCam 335R - Roll 2	617.946	1	1386	High Contrast Imaging of Exoplanets and Exoplanetary Systems with JWST	Hinkley, Sasha
V01184015001	2022-09-06 00:00	F356W	MASKA335R	LP 944-20	LP 944-20 roll 1	846.844	1	1184	A NIRCam Coronagraphic Imaging Survey of Nearby Young M Dwarfs	Schlieder, Joshua
V01184016001	2022-09-06 01:43	F356W	MASKA335R	LP 944-20	LP 944-20 roll 2	846.844	1	1184	A NIRCam Coronagraphic Imaging Survey of Nearby Young M Dwarfs	Schlieder, Joshua

The columns in the above provide information on the observation overall (visit number, date and time, instrument settings, etc), the target and observation label, the duration in seconds and number of dither points used, and which APT program these data are from.

If You Want to Download Public Data without API TOKEN

- first query data with spaceKLIP
- then use *astroquery.mast*, check [tutorial on my github](#)

Download public files without TOKEN

```
[5]: for row in table:
      filename = str(row["filename"])
      uri = f"mast:JWST/product/{filename}"
      local_path = uncal_dir / filename

      if local_path.exists():
          print(f"Already exists: {local_path}")
          continue

      print(f"Downloading {filename}")
      result = Observations.download_file(
          uri,
          local_path=str(local_path),
          cache=False,
      )
      print(result)
```

```
Downloading jw01386001001_0310a_00007_nrcalong_uncal.fits
Downloading URL https://mast.stsci.edu/api/v0.1/Download/file?uri=mast:JWST/product/jw01386001001_0310a_00007_nrcalong_uncal.fits to data_
nircam_hd65426/uncal/jw01386001001_0310a_00007_nrcalong_uncal.fits ... [Done]
('COMPLETE', None, None)
Downloading jw01386001001_0310a_00008_nrcalong_uncal.fits
Downloading URL https://mast.stsci.edu/api/v0.1/Download/file?uri=mast:JWST/product/jw01386001001_0310a_00008_nrcalong_uncal.fits to data_
nircam_hd65426/uncal/jw01386001001_0310a_00008_nrcalong_uncal.fits ... [Done]
('COMPLETE', None, None)
Downloading jw01386001001_0310a_00009_nrcalong_uncal.fits
Downloading URL https://mast.stsci.edu/api/v0.1/Download/file?uri=mast:JWST/product/jw01386001001_0310a_00009_nrcalong_uncal.fits to data_
nircam_hd65426/uncal/jw01386001001_0310a_00009_nrcalong_uncal.fits ... [Done]
('COMPLETE', None, None)
```

Stage 3 Reductions: KLIP

PSF Subtraction: Option Using pyKLIP

Now that we have cleaned up our data to account for alignment and bad pixels, we are ready to perform PSF subtraction. SpaceKLIP supports several algorithms for this step, including the recommended `pyKLIP` and the JWST `Coron3Pipeline`. In this case, we will use pyKLIP for subtraction.

You can customize the PSF subtraction process by adjusting the following settings:

- `mode`: Choose from different image processing techniques, such as ADI, RDI, or a combination of both.
- `annuli`: This parameter determines the number of concentric ring-shaped regions used for PSF subtraction. By specifying different numbers of annuli, you control the radial regions where the subtraction is applied, helping to remove the PSF at various distances from the center of the star. Default is [1].
- `subsections`: This parameter defines the number of smaller regions or segments within each annulus where the PSF subtraction is applied. By breaking down the annuli into subsections, the algorithm can fine-tune the subtraction process. The default value is [1].
- Number of KL Modes (`numbasis`): This parameter specifies how many Principal Component Analysis (PCA) modes are used to build the PSF model for subtraction. PCA modes capture different patterns and features in the data, allowing the model to represent and subtract the PSF more accurately. The default values are [1, 2, 5, 10, 20, 50, 100], which allows us to test different PSF models.
- `algo`: Select the processing algorithm (here, klip).
- `save_rolls`: A roll refers to a set of images taken during a specific pointing direction or orientation of the telescope. Enabling this parameter will save the PSF-subtracted versions of each individual science roll separately, in addition to the roll-combined final product.

Refereed JWST/NIRCam and JWST/MIRI Coronagraph Literature (1/5)

Paper title	Authors	Target	Filters
JWST/MIRI coronagraphic performances as measured on-sky	Boccaletti et al. (2022)	Single stars + reference stars	F1065C, F1140C, F1550C, F2300C
The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems I: High-Contrast Imaging of the Exoplanet HIP 65426 b from 2 to 16 μm	Carter et al. (2023)	HIP 65426 b	NIRCam: F250M, F300M, F356W, F410M, F444W; MIRI: F1140C, F1550C
First Observations of the Brown Dwarf HD 19467 B with JWST	Greenbaum et al. (2023)	HD 19467 B	6 filters spanning 2.5–4.6 μm ; exact names not specified
JWST/NIRCam Coronagraphy of the Young Planet-hosting Debris Disk AU Microscopii	Lawson et al. (2023)	AU Mic disk; companion search	F356W, F444W
Searching for Planets Orbiting Fomalhaut with JWST/NIRCam	Ygouf et al. (2024)	Fomalhaut (α PsA)	F356W, F444W
JWST-TST High Contrast: Asymmetries, Dust Populations, and Hints of a Collision in the β Pictoris Disk with NIRCam and MIRI	Rebollido et al. (2024)	β Pictoris disk	8 filters spanning 1.8–23 μm ; exact list not specified

Listed by ChatGPT, so the list may have incorrect information or miss some papers.

Some data are possibly still in proprietary period – data are usually public a year after observations.

Refereed JWST/NIRCam and JWST/MIRI Coronagraph Literature (2/5)

Paper title	Authors	Target	Filters
JWST/NIRCam Detection of the Fomalhaut C Debris Disk in Scattered Light	Lawson et al. (2024)	Fomalhaut C disk	3.6 and 4.4 μm bands
JWST/NIRCam observations of the young giant planet β Pic b	Kammerer et al. (2024)	β Pic b	6 filters covering $\sim 1.7\text{--}5\text{ }\mu\text{m}$; exact names not specified
Imaging detection of the inner dust belt and the four exoplanets in the HR 8799 system with JWST's MIRI coronagraph	Boccaletti et al. (2024)	HR 8799 inner belt + planets b, c, d, e	MIRI
Unveiling the HD 95086 system at mid-infrared wavelengths with JWST/MIRI	Mâlin et al. (2024)	HD 95086 b + inner / outer disk components	F1065C, F1140C, F2300C
JWST/NIRCam 4–5 μm Imaging of the Giant Planet AF Lep b	Franson et al. (2024)	AF Lep b	F444W explicitly; other exact names not specified
A temperate super-Jupiter imaged with JWST in the mid-infrared	Matthews et al. (2024)	ϵ Indi Ab	Two narrowband filters at 10.65 and 15.50 μm

Refereed JWST/NIRCam and JWST/MIRI Coronagraph Literature (3/5)

Paper title	Authors	Target	Filters
First unambiguous detection of ammonia in the atmosphere of a planetary mass companion with JWST/MIRI coronagraphs	Mâlin et al. (2025)	GJ 504 b	MIRI
MIRI-JWST mid-infrared direct imaging of the debris disk of HD 106906: Structure and mass of the disk	Rouan et al. (2025)	HD 106906 disk	11.3 and 15.5 μm
Evidence for a sub-Jovian planet in the young TWA 7 disk	Lagrange et al. (2025)	TWA 7; candidate TWA 7 b	MIRI
Follow-up Exploration of the TWA 7 Planet–Disk System with JWST NIRCam	Crotts et al. (2025)	TWA 7 disk + candidate companion(s)	F200W, F444W
Serendipitous observation of a white dwarf companion to a JWST/MIRI coronagraphic calibrator	Venner, Limbach, Mâlin, Blouin, Boccaletti, and Pearce (2025)	HD 218261 B in calibrator data for HR 8799 work	10–16 μm range
Probing the Outskirts of M Dwarf Planetary Systems with a Cycle 1 JWST NIRCam Coronagraphy Survey	Bogat et al. (2025)	AU Mic; HIP 17695; TYC 5899; G 7-34; Fomalhaut C; AP Col; 2M J0944; LP 944-20; 2M J0443+0002	F356W, F444W

Refereed JWST/NIRCam and JWST/MIRI Coronagraph Literature (4/5)

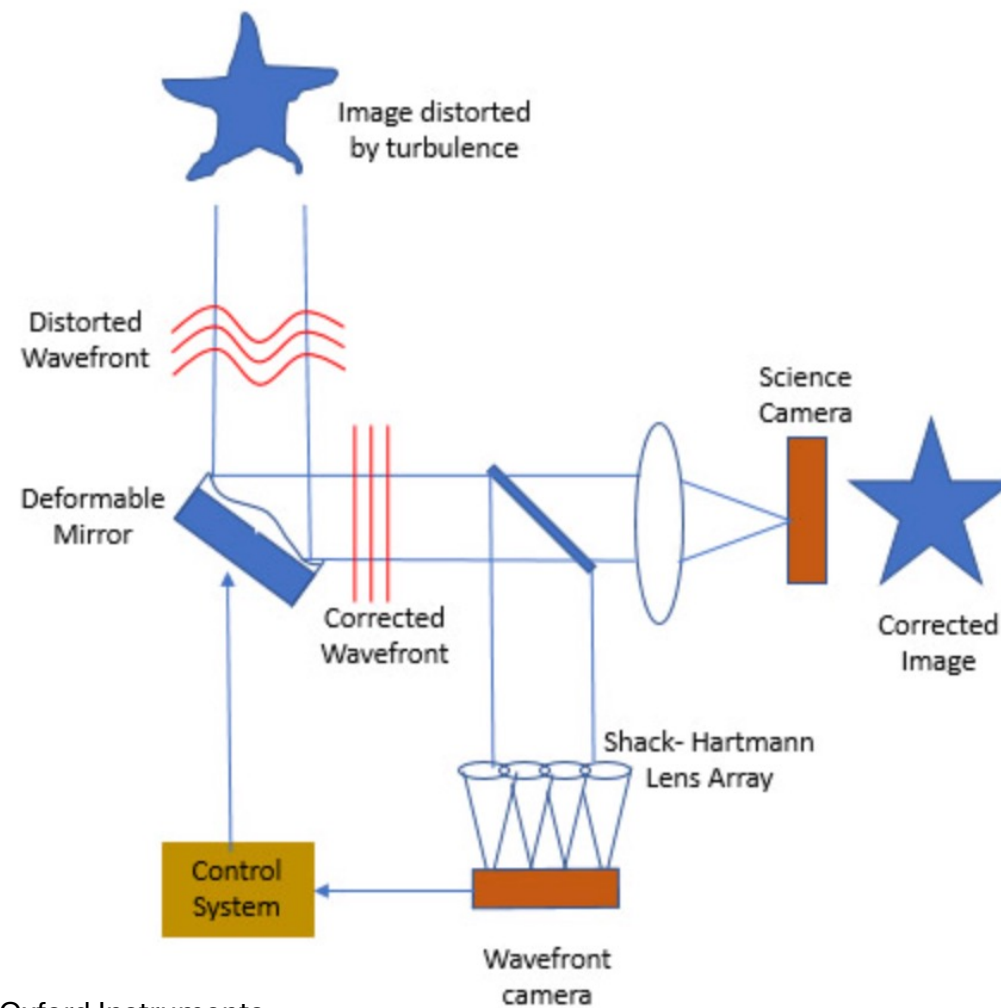
Paper title	Authors	Target	Filters
Searching for Planets Orbiting ϵ Eridani with JWST/NIRCam	Llop-Sayson et al. (2025)	ϵ Eridani; inner RV planet + outer ring-sculpting planet search	F356W, F444W
Direct Measurement of Extinction in a Planet-hosting Gap	Cugno et al. (2025)	AS 209 / background star AS 209bkg	NIRCam exact names not specified
JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet in a Dynamically Hot, Multi-planet System	Bardalez Gagliuffi et al. (2025)	14 Her c	F200W, F444W
Worlds Next Door: A Candidate Giant Planet Imaged in the Habitable Zone of α Centauri A. I. Observations, Orbital and Physical Properties, and Exozodi Upper Limits	Beichman et al. (2025)	α Cen A; candidate planet	F1550C
Worlds Next Door: A Candidate Giant Planet Imaged in the Habitable Zone of α Centauri A. II. Binary Star Modeling, Planet and Exozodi Search, and Sensitivity Analysis	Sanghi et al. (2025)	α Cen A (+ binary-star light from α Cen B)	F1550C

Refereed JWST/NIRCam and JWST/MIRI Coronagraph Literature (5/5)

Paper title	Authors	Target	Filters
JWST/NIRCam Coronagraphic Search for Hidden Planets in the HD 163296 Protoplanetary Disk	Uyama et al. (2025)	HD 163296	F410M, F200W
Discovery of a Debris Disk Around TWA 20	Palatnick et al. (2025)	TWA 20 disk	F200W, F444W
JWST/NIRCam observations of HD 92945 debris disk: An asymmetric disk with a gap	Lazzoni et al. (2025)	HD 92945 disk	F200W, F444W
JWST/MIRI coronagraphic search for planets in systems with gapped exoKuiper belts and proper motion anomalies	Bendahan-West et al. (2026)	HD 92945; HD 107146; HD 206893	F1140C (11.4 μ m)

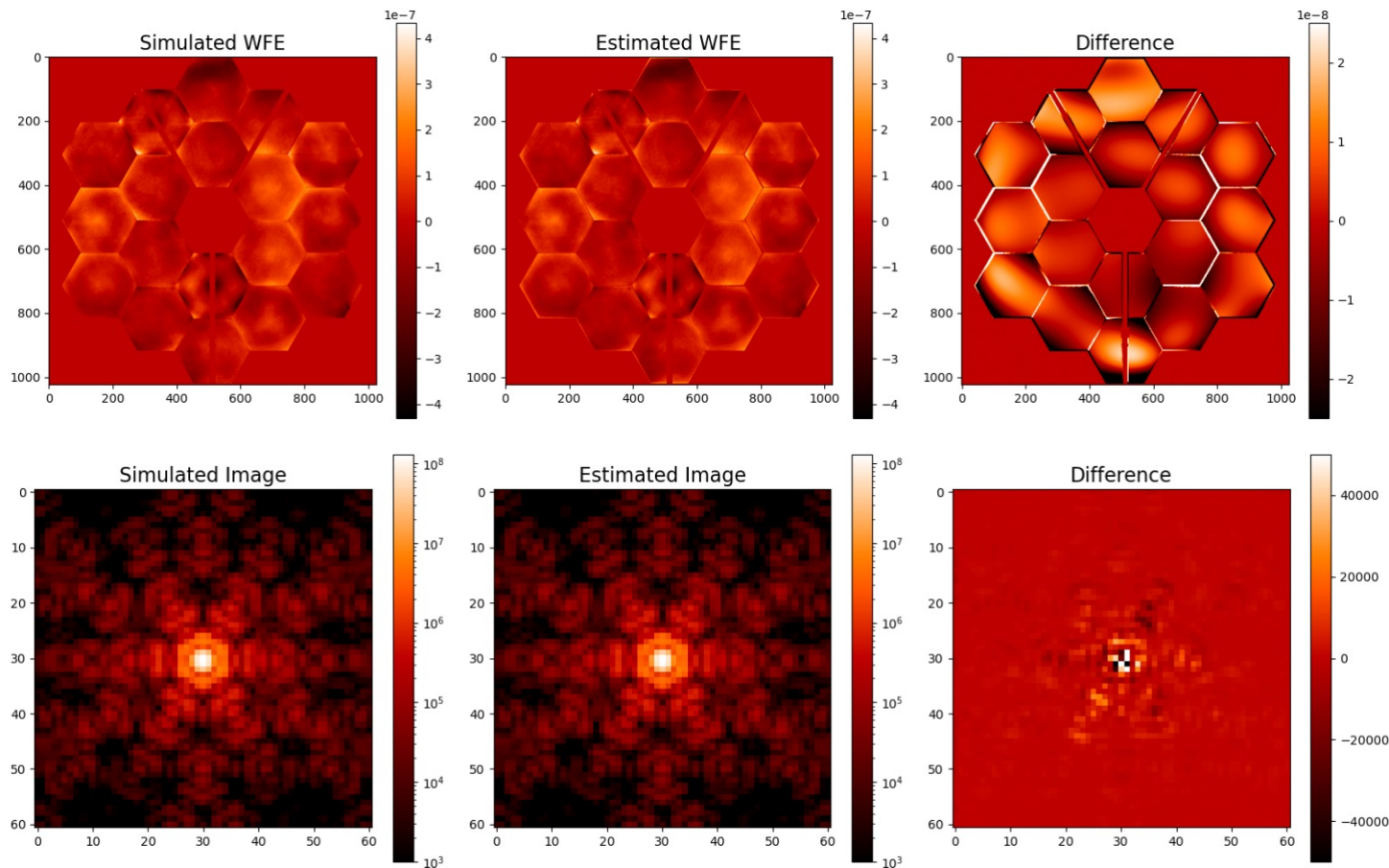
Auxiliary Slides

Adaptive Optics Technique



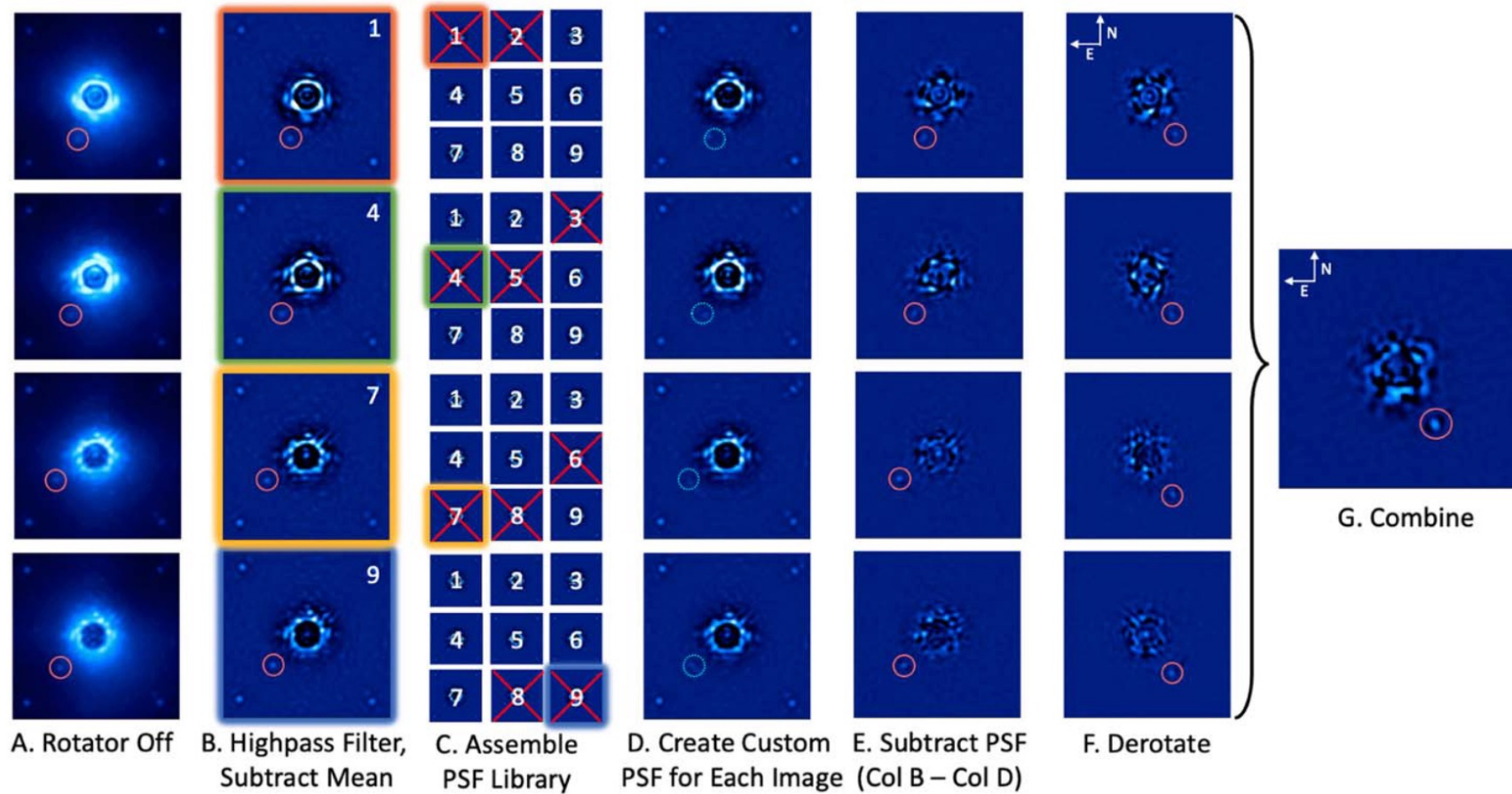
Difficulty in Simulating Instrumental PSF with STPSF

- PSF pattern depends on configurations, spectral type of the source, WFEs
- STPSF warps the configurations and WFEs, but cannot fully reproduce the observed PSF



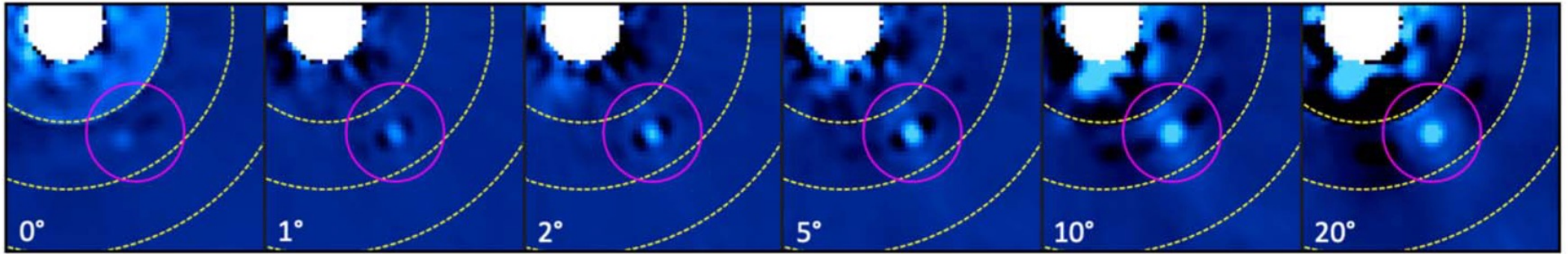
KLIP/PCA-based PSF Library Assembly

Karhounen–Loeve Image Processing/Principal Component Analysis

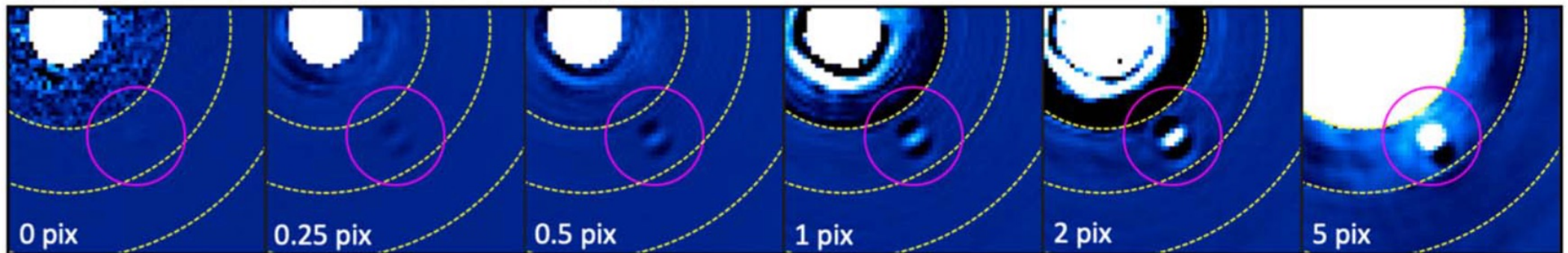


Self-subtraction Effect and its Dependence

Azimuthal Self-Subtraction (Angular Differential Imaging)



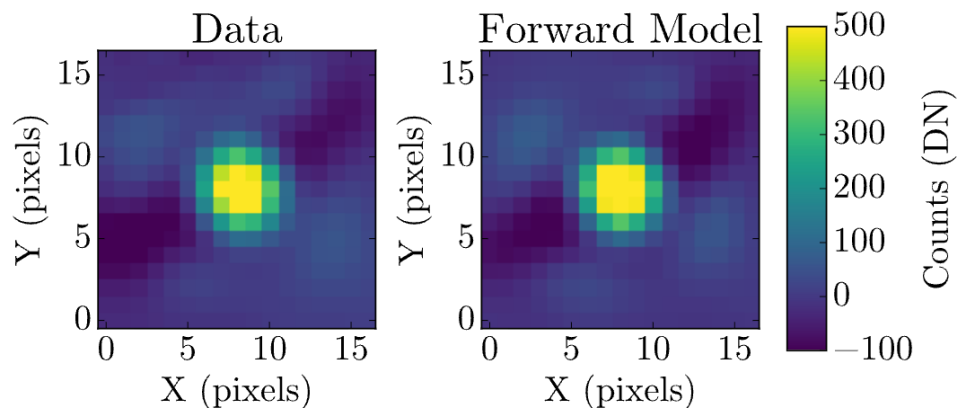
Radial Self-Subtraction (Spectral Differential Imaging)



Follette et al. 2023

Forward Modeling

- By injecting fake sources and rerunning the pipeline, one can evaluate how real sources (companions, disks) are over-subtracted via post-processing, particularly ADI or SDI.
- This helps to correct ‘throughput’ of post-processing when calculating contrast limits and extracting photometry and astrometry.



Left: point-source case (Wang et al. 2016)
Right: disk case (Mazoyer et al. 2020)

